

Searching:

There are several applications in which we store a high amount of data and search for a piece of information within it.

For example, we may be searching for a particular phone number in a telephone directory which is stored in some order.

Basic search Techniques:

Some Terminology: A file is a collection of records. Each record intern is a group of different data items. There is a key associated with each record, which is used to distinguish among different records.

An internal key is a field of information within a record at a particular position from the beginning of the record.

In some cases, a separate table of keys are maintained which are having pointers to the records. This type of key is called external key.

For every file there is at least one set of keys that is unique (no two records have the same value). Such a key is called primary key.

Example: Register number is the primary key for the student information system.

In some cases, the key will not be unique since there may be two records with the same key in the file. Such a key is called secondary key.

Example: Name is the secondary key for the student information system.

A search algorithm is an algorithm tries to find a record whose key is given. The algorithm may return the entire record or a pointer to that record. A successful search is often called a retrieval.

The table of records in which key is used for retrieval is called a search table or dictionary.

Searches in which the entire table is constantly in the main memory are called internal searches.

Searches in which the entire table is kept in the auxiliary memory are called external searches.